Fiscal Year:	FY 2026	**Task Last Updated:**	FY 07/24/2025
PI Name:	Bell, Suzanne Ph.D.		
Project Title:	Human Factors Behavioral Performance Risk Characterization Research for Artemis II		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/03/2022	**End Date:**	09/30/2027
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:	Artemis II		
Flight Assignment:	NOTE: End date changed to 09/30/2027 per L. Juliette/HFBP (Ed., 8/7/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Vera, Alonso Ph.D. (NASA Ames Research Center) Flynn-Evans, Erin Ph.D. (NASA Ames Research Center) Young, Millennia Ph.D. (NASA Johnson Space Center) Dev, Sheena Ph.D. (KBR/NASA Johnson Space Center) Landon, Lauren Ph.D. (KBR/NASA Johnson Space Center) Anderson, Steven (KBR/NASA Johnson Space Center) Garrett, James (NASA Johnson Space Center) Begerowski, Sydney (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Artemis II will mark the first time NASA astronauts go beyond low-Earth orbit (LEO) since the Apollo era, and the first astronauts heading into space in the Orion vehicle. As such, it provides a critical opportunity to refine our understanding of the likelihood and consequences associated with the Behavioral Medicine (BMed), Team, Human System Integration Architecture (HSIA), and Sleep Risks, and prepare for future Moon and Mars missions. The overarching goal of our research is to utilize Artemis II data to further define the likelihood and consequences of these risks, and to create a research infrastructure that can be expanded to include future Artemis missions. We propose a 3-phase research effort. In Phase I, we will use summaries of existing research (e.g., evidence books and Directed Acyclic Graphs/DAGs), literature reviews, and subject matter expert (SME) input to identify and conceptualize key performance metrics, contributing factors, and BMed, Team, HSIA, and Sleep risk constructs related to performance decrements. We will describe how the constructs are expected to arise in spaceflight, and conduct a review that summarizes how metrics (e.g., behaviors and trace data, words and linguistics, and physiological data) derived from data streams available in Artemis II (e.g., audiovisual data) can serve as indicators of these constructs for in-mission measurement during Artemis. The Phase I effort will result in a finalized pre- and post-mission protocol for Artemis II, along with a measurement and coding scheme for in-mission Artemis II data. Phase II includes data collection from the Artemis II mission. Phase III will include data processing, data analysis under a Bayesian framework, coding, depiction, analysis, and report writing.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. Artemis II provides a critical opportunity to refine our understanding of the likelihood and consequences associated with BMed, Team, HSIA, and Sleep Risks, and prepare for future Moon and Mars missions. Two significant constraints shape the research methodology. First, there is currently no in-mission crew time available to complete measures. In-mission data will need to be collected unobtrusively from available data streams (e.g., audiovisual, existing records such as schedules, and actigraphy). Second, Artemis II is anticipated to be a crew of 4 astronauts in flight for 10 days followed by additional crewed Artemis missions. This creates a scenario where there is important data for understanding risk characterization; however, there will be a small sample size from Artemis II, and risk characterization will need to be updated over time as additional crews fly in later Artemis missions.
Research Impact/Earth Benefits:	Our research program and associated deliverables will substantially reduce NASA’s Human Research Program Behavioral Medicine (BMed), Team, Human Systems Integration Architecture (HSIA), and Sleep Risks and inform gaps in knowledge related to future Moon and Mars missions. Our primary contributions will be in advancing less obtrusive methodologies for monitoring and measuring risk metrics; examining potential BMed, Team, HSIA, and Sleep risks in a new vehicle, with new mission scenarios and procedures; and characterizing risks beyond LEO as multiple stressors converge. The advances in less obtrusive methodologies may have broader application to other private and public sector organizations interested in team dynamics, well-being, sleep, and human factors. Given that Artemis II will be the first human spaceflight in Orion, these data will be particularly valuable for informing modifications necessary for future Moon and Mars missions. Finally, Artemis II data will be the first systematic data collection related to these risks beyond LEO and allow for the potentially additive or synergistic effect of multiple stressors on sleep, and psychological and team health as the crew interacts with new vehicle and system interface deigns, as well as mission scenarios.
Task Progress:	Over the past year, the Behavioral Health and Performance Laboratory and collaborators from the Fatigue Laboratory and Human Systems Integration Architecture (HSIA) refined research products and protocols completed under Phase 1. The research team identified key constructs pertaining to the BHealth, Teams, Sleep, and HSIA risk areas likely to be most relevant to Artemis II mission. The team consulted the literature as well as SMEs to refine the research protocol given changes in available data streams (e.g., removal of audio-visual and physiological data for use in research). Changes were made to surveys and post-mission crew debriefs in order to ensure comprehensive coverage of in-mission phenomena retrospectively. Alongside these efforts, the research team successfully completed the crew informed consent briefing and continue to support the informed consent process with the adapted protocol. We also are preparing for data collection.
Bibliography Type:	Description: (Last Updated: 08/01/2025)
Abstracts for Journals and Proceedings	Begerowski SR, Bell ST, Mathieu JR, Gorman J, Gupta P, Rosopa PR, Pool RN, Rosopa EH, Chaspari T. "Under the radar: Measuring team dynamics without disturbance [IGNITE]." Society for Industrial Organizational Psychology Annual Conference, Denver, Colorado, April 2-5, 2025. Abstracts. Society for Industrial Organizational Psychology Annual Conference, Denver, Colorado, April 2-5, 2025. , Apr-2025